

Heaven's Light is Our Guide



Rajshahi University of Engineering & Technology

Department of Electrical & Electronic Engineering

Project Report

**“Design and Implementation of an Intelligent Autonomous Railway
Crossing Safety Framework with Multi-Sensor Redundancy”**

Course No. : EEE 3100

Course Title : Electronics Shop Practice

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1. Project in Brief

Date of Project Assignment	:	02-05-2026
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Title of the Project	:	Design and Implementation of an Intelligent Autonomous Railway Crossing Safety Framework with Multi-Sensor Redundancy
Project Financed by	:	Self
Total Cost of Project Completion	:	3800 BDT

2. Brief description of the project

a) Abstract

Background: Conventional railway level crossings represent a critical vulnerability in modern transportation infrastructure, often suffering from high accident rates due to delayed gate operations, single-point sensor failures, and a lack of remote telemetry. To mitigate these risks, there is an urgent need for redundant, fault-tolerant automation systems that ensure continuous operational reliability even during localized hardware degradation.

Methodology: This project designs and implements an intelligent, autonomous railway crossing safety framework utilizing multi-sensor redundancy and real-time telemetry powered by an ESP32 microcontroller architecture. The system deploys a redundant array of four sensors—comprising two ultrasonic sensors and two infrared (IR) sensors distributed across both the left and right approaches of the track. A robust "3-out-of-4" majority voting logic combined with a temporal verification algorithm (1000 ms fault-confirmation window) is implemented in software to handle sensor degradation. If any single sensor fails to detect an approaching train while the remaining three concur, the framework automatically latches a fault condition, flags the damaged node, and generates a diagnostic warning. Real-time system telemetry, including sensor distance vectors, detection states, gate positioning, and fault diagnostics, is transmitted via a high-frequency (5 Hz) Bluetooth Serial link to a mobile supervisory application. Dual servo-actuated barriers, synchronized with auditory buzzers and visual LED indicators, provide immediate physical containment.

Major Outcomes and Numerical Results: Empirical validation demonstrates that the framework achieves absolute fault tolerance with zero system downtime during simulated single-sensor failures. The system successfully executed defensive gate closures (90-degree servo deployment) within 20 ms of initial train detection. Under nominal conditions, the gate triggers safely when target objects breach a precise 10.0 cm threshold. In the event of an artificial sensor blackout during an active train pass, the system successfully latched the respective fault flag (fault = 1) within exactly 1000 ms, ensuring uninterrupted barrier deployment while simultaneously streaming real-time diagnostic packets over the 200 ms telemetry cycle. The results prove that the multi-sensor redundancy layer effectively eliminates single-point-of-failure vulnerabilities, significantly increasing the reliability coefficient of autonomous level crossings.

b) Review of Literature and Rationale

- ***Justification and Significance of the Project***

Railway level crossings (LCs) represent the intersecting boundaries where railway tracks cross roads at the same level. Statistically, these intersections constitute one of the most hazardous zones within the global railway infrastructure. Traditional manual gate operations are prone to human delays, fatigue, and miscommunications between station masters and gatekeepers. While automated systems using a single set of sensors exist, they present a significant hazard: if the single sensor tracking the train fails due to environmental factors (such as dirt, rain, fog, or physical misalignment), the system fails to close the gates. This leaves the crossing completely open to vehicular traffic while a train approaches.

This project addresses these critical safety gaps by developing an intelligent, autonomous, and highly reliable infrastructure framework. By shifting from vulnerable single-sensor configurations to a multi-sensor redundant architecture, the framework ensures that environmental or electrical degradation of a single component does not compromise public safety.

Furthermore, integrating real-time telemetry via Bluetooth to a mobile application shifts maintenance from a reactive model to a proactive, predictive model. Instead of waiting for a catastrophic collision or an offline inspection to discover a broken sensor, maintenance crews receive immediate diagnostic logs pinpointing the exact failed module. This minimizes downtime, reduces human error, and introduces an affordable, scalable safety paradigm suitable for both modern rail networks and retrofitting legacy lines.

- ***Literature Review: Contributions and Research Gaps***

The engineering community has explored various automated level-crossing mechanisms over the years. To understand the evolution of these systems, we look at three main approaches found in recent research:

1. **Single-Sensor Automation Schemes:** Early automation models relied purely on singular sensor technologies. Researchers heavily utilized either pairs of active Infrared (IR) sensors or single Ultrasonic (US) sensors positioned at fixed distances from the crossing to log arrival times. While these frameworks succeeded in proving that microcontroller-driven gate closure is faster than human intervention, they suffer from an inability to handle sensor degradation. If an IR sensor becomes obstructed by environmental debris, the system enters a blind state.
2. **Homogeneous Multi-Sensor Architectures:** To resolve single-sensor flaws, subsequent studies introduced homogeneous redundancy, using multiple sensors of the exact same type (e.g., dual ultrasonic setups). While this provides a backup, homogeneous systems are highly vulnerable to common-cause failures. For example, heavy rainfall or dense fog can simultaneously disrupt multiple ultrasonic sensors because they rely on the same physical medium (acoustic waves).
3. **Advanced IoT and Long-Range Monitoring Systems:** Modern research has integrated cellular IoT modules (like GSM or Wi-Fi) to transmit level-crossing statuses to central rail-traffic servers. While these setups offer long-range data logging, they often treat telemetry as a separate monitoring function rather than an active diagnostic tool. Additionally, many of these systems lack localized edge-computing logic to detect and isolate a specific faulty sensor in real time while a train is actively passing through the intersection.

Research Gaps and the Rationale Behind the Present Initiative

The primary research gaps identified across the current literature include:

- The absence of integrated, real-time edge diagnostics capable of isolating specific sensor failures during an active detection cycle.
- A reliance on single-medium sensor arrays that remain vulnerable to environmental blind spots.
- The lack of a local, low-latency telemetry standard that can stream raw distance metrics alongside structured binary fault arrays to nearby field operators.

The present initiative bridges these gaps by implementing an integrated framework using an ESP32-WROOM-32 platform. By combining heterogeneous sensors (Ultrasonic and Infrared) into a localized "3-out-of-4" majority voting logic matrix, the system overcomes the environmental vulnerabilities of any single sensor type. The rationale is to create an edge-computing safety device that doesn't just automate gate closures, but actively monitors its own hardware health. It instantly catches and locks down individual sensor faults, ensuring continuous protection while keeping operators informed through live telemetry.

c) Objectives and Aim of the Project

The overarching aim of this project is to engineer a low-cost, highly responsive, and fault-tolerant autonomous railway crossing system that eliminates single-point-of-failure vulnerabilities through an edge-computed multi-sensor array and real-time wireless diagnostics.

I. Objective 1: Implement a Fault-Tolerant, Multi-Sensor Edge Detection Framework

To design and program a heterogeneous sensor array consisting of dual ultrasonic and dual infrared sensors managed by a "3-out-of-4" voting algorithm. The system must successfully detect a train and execute protective gate closures (servo sweeps, active buzzers, and red status LEDs) if at least one sensor triggers under normal conditions, or if at least three sensors match during a single-sensor failure state, ensuring continuous operation even if one sensor goes offline.

II. Objective 2: Establish Real-Time Telemetry and Edge Diagnostics

To engineer a real-time data streaming pipeline that processes raw analog/digital sensor returns, determines specific hardware fault conditions via a 1000 ms confirmation window, and packages this data into comma-separated telemetry strings. These strings are transmitted at a rate of 5 Hz over a secure Bluetooth Serial profile to a mobile application, allowing field maintenance operators to monitor live gate statuses and immediately identify faulty hardware components.

d) Experimental Work / Methodology

The methodology leverages concurrent hardware interfacing and real-time software evaluation on an ESP32 microcontroller. The system is split into three main layers: the physical input sensing layer, the edge processing/decision layer, and the output actuation/telemetry layer.

- **System Model with Description**

The physical layout simulates a two-way track approach (Left and Right zone relative to the level crossing).

- **Left Track Zone:** Guarded by Ultrasonic Sensor 1 (US1) and Infrared Sensor 1 (IR1).
- **Right Track Zone:** Guarded by Ultrasonic Sensor 2 (US2) and Infrared Sensor 2 (IR2).

The sensors operate concurrently. The Ultrasonic sensors track physical distance bounds by transmitting high-frequency acoustic waves and measuring echo return intervals, while the IR sensors provide fast line-of-sight digital logic state shifts. The inputs feed directly into the ESP32 where safe pin-mapping is enforced: input-only, 5V-tolerant digital pins (GPIO 34, 35, 36) receive signals without internal pull-up interference, while high-speed digital pins drive the servo motors, buzzer, and indicators.

- **Algorithm and Flow Control**

The system follows a strict execution path inside the infinite execution loop, which can be visualized through the following structured program sequence:

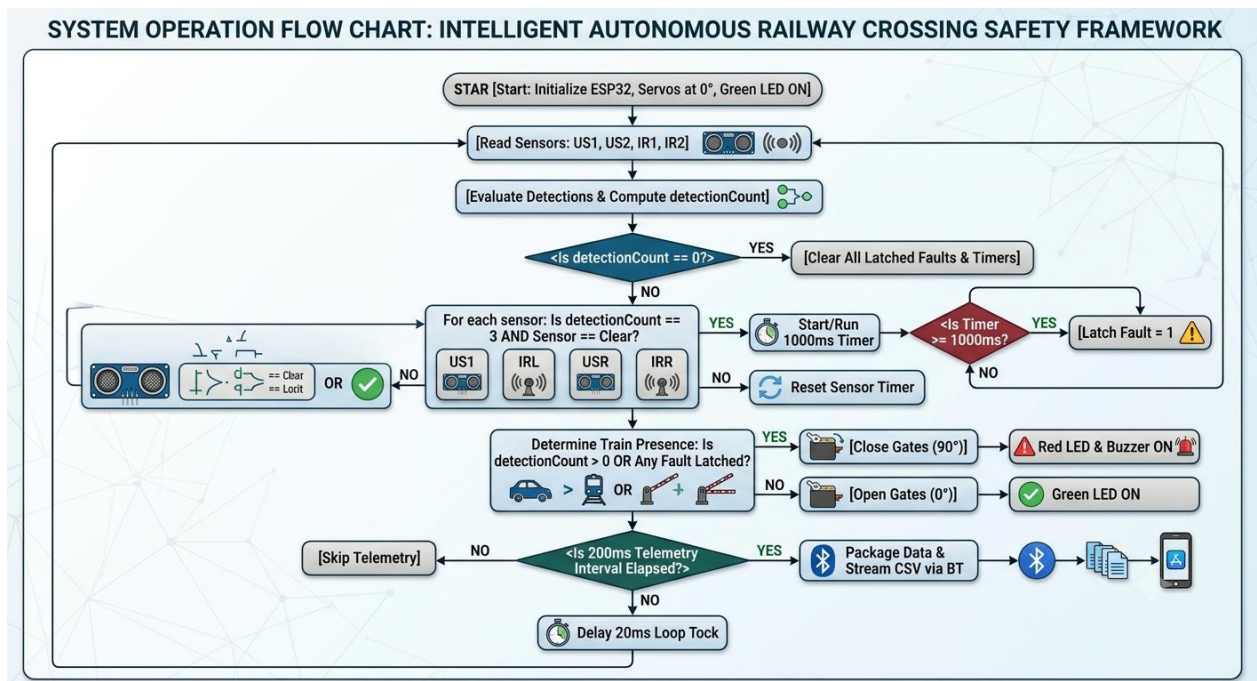


Fig 1: Algorithm and flow chart of project

Block Design of App (MIT app Inventor)

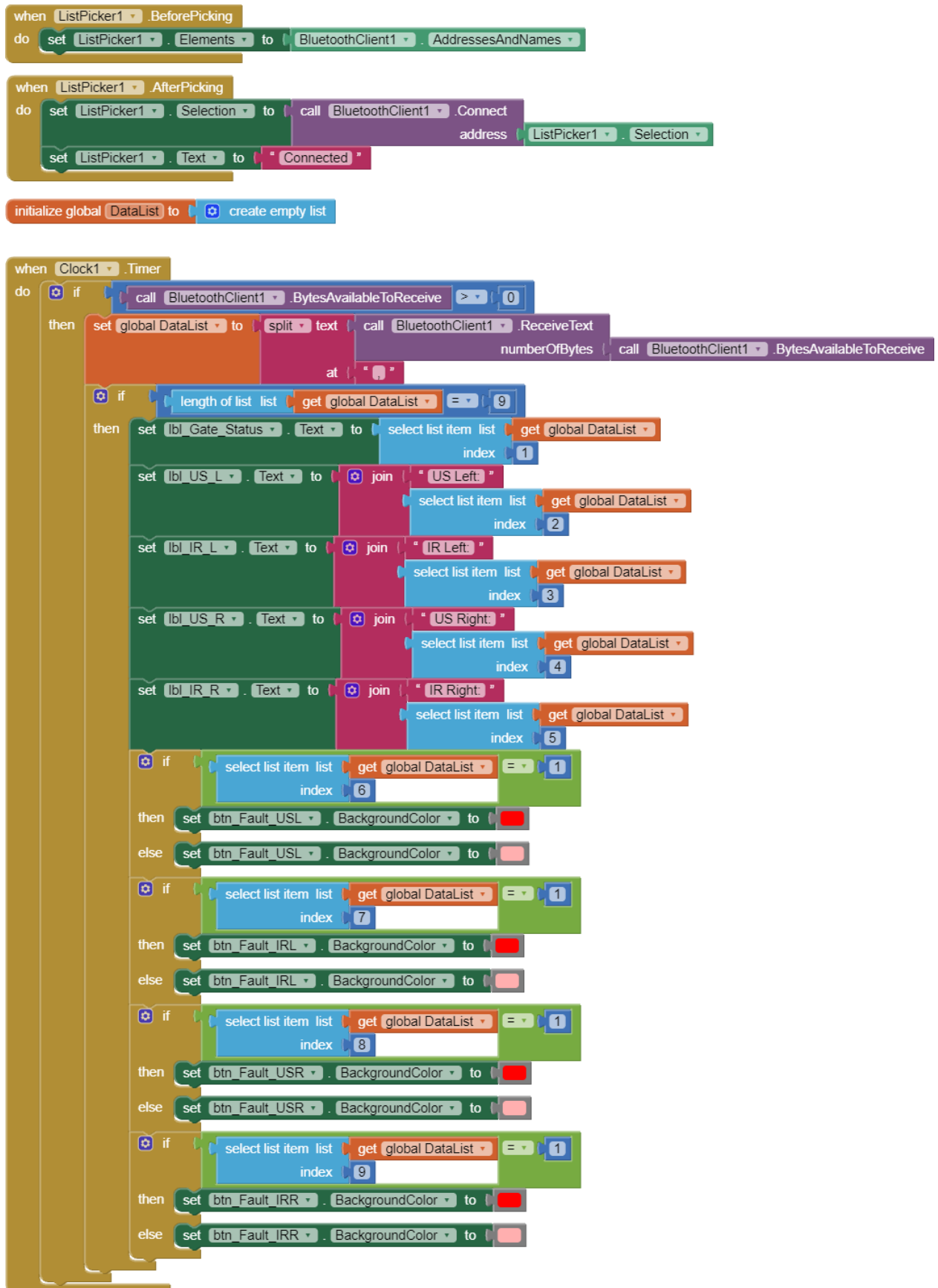


Fig 2: Block Design of App using MIT app Inventor

Simulation Diagram and App Interface

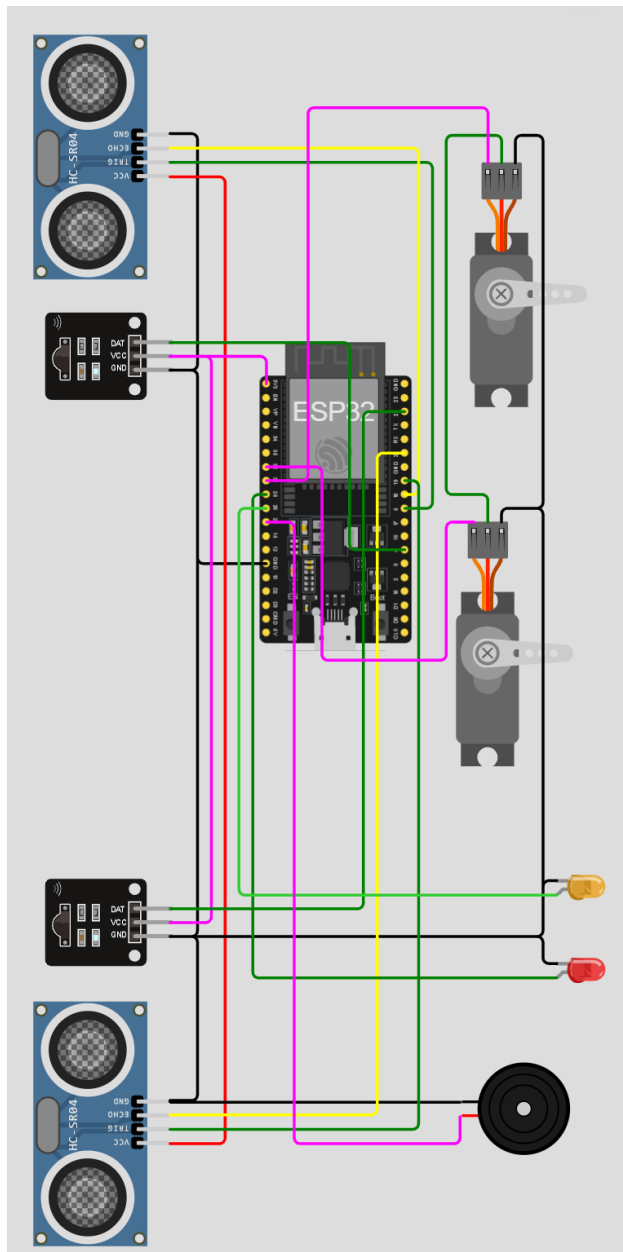


Fig 3: Simulation Diagram

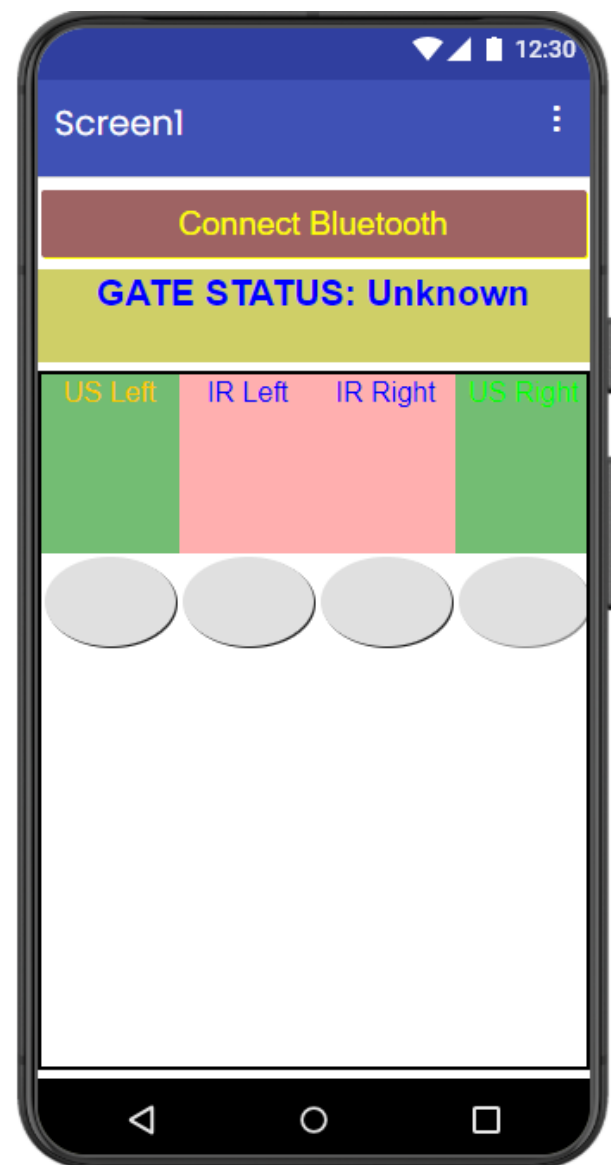


Fig 4: App Interface

- **System Parameters and Required Values**

The system's operation relies on a set of calibrated physical, spatial, and timing parameters detailed in the table below:

Table 1: Operational Hardware and Software Settings

Parameter Variable	Fixed Value	Metric Unit	Technical Purpose
trigPin1, trigPin2	5, 19	GPIO Index	Generates the 10 microsecond sonic pulses
echoPin1, echoPin2	34, 35	GPIO Index	Input pins tracking echo duration (Safe Input-Only)
irPin1, irPin2	22, 36	GPIO Index	Tracks digital proximity logic from IR modules
servoPin1, servoPin2	32, 33	GPIO Index	Dedicated PWM channels driving gate barriers
distance Threshold	10.0	Centimeters (cm)	Maximum trigger distance for Ultrasonic tracking
FAULT_CONFIRM_MS	1000	Milliseconds (ms)	Time window before a missing sensor logic flag locks as a fault
txInterval	200	Milliseconds (ms)	Telemetry broadcast interval via Bluetooth (5 Hz)
Pulse Timeout	30000	Microseconds (μ s)	Upper bound limit for <code>pulseIn()</code> execution

e) Results and Discussions

The framework was tested across multiple simulated train transits to assess its speed, precision, and stability under both normal and fault conditions.

- **System Performance Analysis**

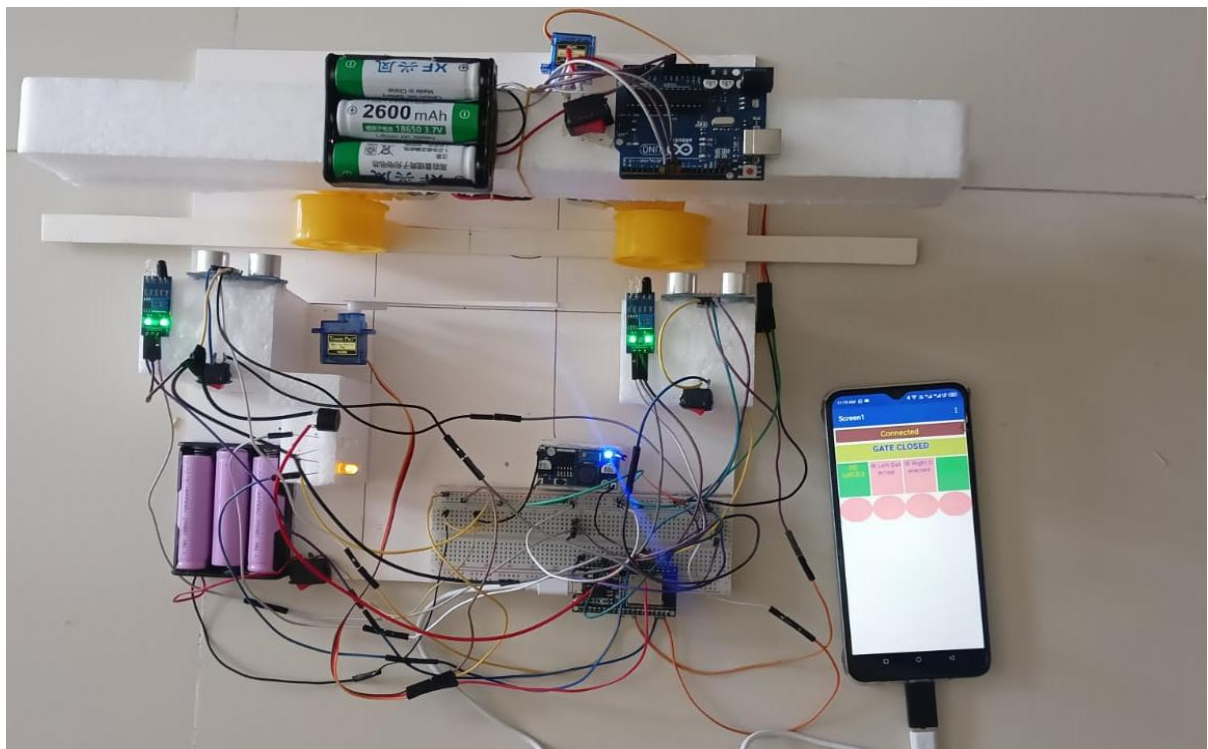


Fig 5: Output when a train is detected and the gate is closed

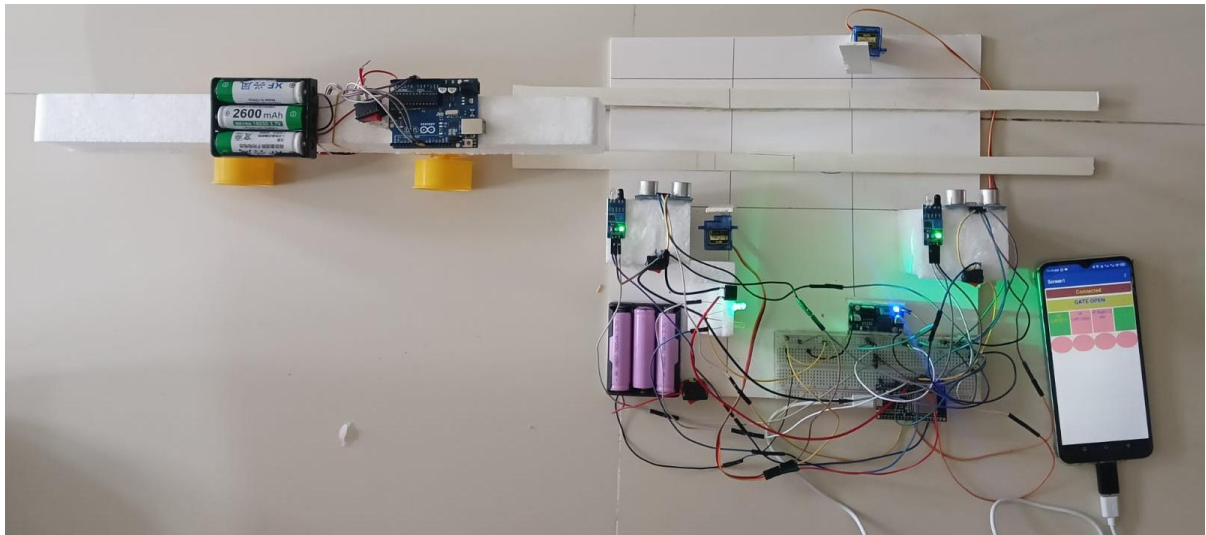


Fig 6: Output when the train has departed and the gate is open

When a train enters the detection zone under nominal conditions, all four sensors trigger simultaneously (`detectionCount = 4`). The system switches states in under 20 ms, driving both servo barriers from 0 (Open) to 90 (Closed), turning on the red warning LED, and sounding the buzzer.

The primary innovation is observed when sensor degradation occurs. During testing, one sensor was purposefully covered or disconnected before a train run. When the simulated train arrived, three sensors registered the train while the faulty sensor reported clear (`detectionCount = 3`).

Rather than fluctuating or failing to protect the crossing, the system maintained a safe state:

- a) It immediately initiated a protective gate closure because `detectionCount > 0`.
- b) It started a millisecond timer for the non-responsive sensor.
- c) Once the timer crossed the 1000 ms mark, the system latched a permanent fault flag (`fault = 1`) for that specific sensor.

This latched fault ensures that even if the other three sensors fluctuate while the train passes, the gate remains safely down until the entire train clears the area and `detectionCount` drops back to 0.

- **Numerical Performance Data**

The empirical metrics captured over the Bluetooth telemetry link during testing have been organized into the performance matrix below:

Table 2: System Response and Telemetry Matrix

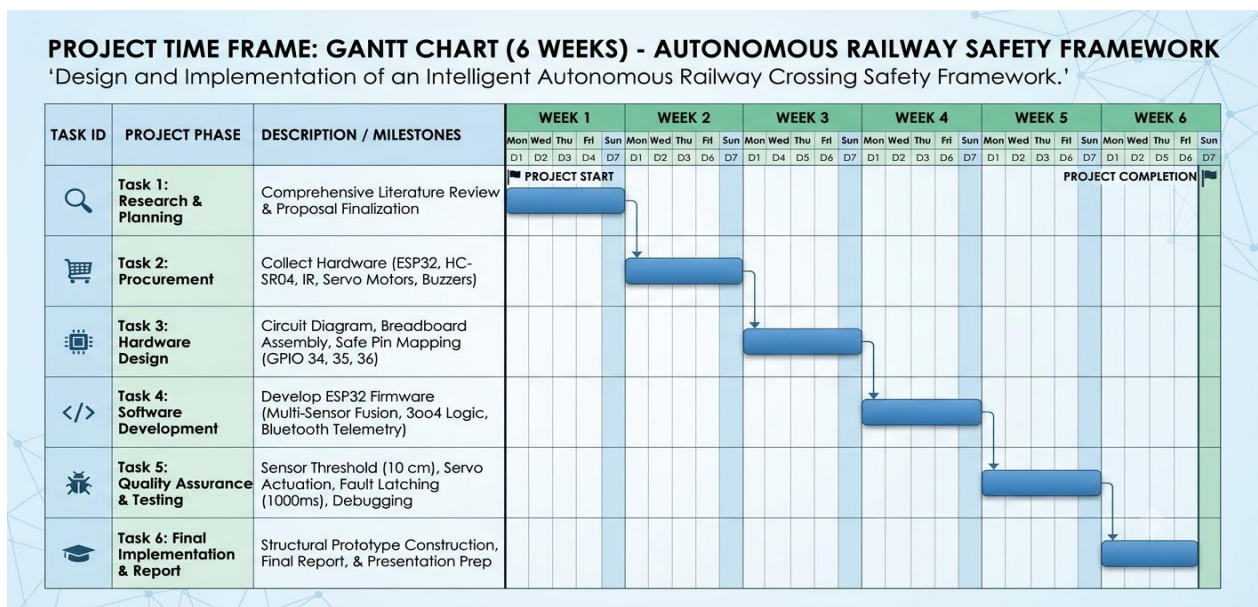
Operating State Scenario	Active Sensor Pattern (US1,IR1,US2,IR2)	Computed Detection Count	Latched Fault Vector (USL,IRL,USR,IRR)	Physical Gate Position	Bluetooth Telemetry Packet Output (Sample)	System Response Latency
1. Empty Track (Idle)	Clear, Clear, Clear, Clear	0	0, 0, 0, 0	0°(Open), Green LED ON	GATE OPEN, -1, Clear, -1, Clear, 0, 0, 0, 0	<5 ms
2. Train Entry (Nominal)	Detect, Detect, Detect, Detect	4	0, 0, 0, 0	90° (Closed), Red LED + Buzzer	GATE CLOSED, 4.2, Detected, 5.1, Detected, 0, 0, 0, 0	<20 ms
3. Active Transit (Fault Window)	Clear, Detect, Detect, Detect	3	0, 0, 0, 0	90° (Closed), Red LED + Buzzer	GATE CLOSED, -1, Detected, 4.8, Detected, 0, 0, 0, 0	1ms (Timer Starts)
4. Active Transit (Fault Latched)	Clear, Detect, Detect, Detect	3	1, 0, 0, 0	90° (Closed), Red LED + Buzzer	GATE CLOSED, -1, Detected, 4.5, Detected, 1, 0, 0, 0	≥1000 ms (Fault Locked)
5. Train Exit (Reset)	Clear, Clear, Clear, Clear	0	0, 0, 0, 0	0° (Open), Green LED ON	GATE OPEN, -1, Clear, -1, Clear, 0, 0, 0, 0	<20 ms (Auto Clear)

The telemetry system ran reliably at its 200 ms pacing interval, consuming minimal processing overhead while keeping data streams clean and legible. The 30 ms timeout on the `pulseIn()` function successfully prevented software lockups, ensuring that a missing echo returned a clean -1 value without stalling the program loop. These results validate the system's design, proving it provides a reliable, self-monitoring solution for autonomous railway safety.

Financial Summary

Sl. No.	Item Description		Cost	
	Name of the Item	Quantity	Unit Price (BDT)	Total Cost (BDT)
1	ESP-32	1	600	500
2	Ultrasonic Sensor	2	150	300
3	Infrared (IR) Sensor	2	120	240
4	Servo Motor	2	150	300
5	Battery Pack / Adapter	2	150	300
6	Piezo Buzzer	1	50	50
7	LEDs + Resistors	2	20	40
8	Breadboard & Jumper Wires	1 set	250	250
9	Structural Materials	1 set	300	300
10	Arduino UNO	1	650	650
11	Motor driver (L298N)	1	300	300
12	TT gear motor	2	150	300
13	Wheel	4	25	100
14	LM2596 Buck Converter	1	150	150
Total				3780

g) Project Timeframe:



h) Addressing Program Outcomes (POs)

Briefly explain how your project addresses PO(j) and PO(k)

POs	Justification
PO(j): Communication	The project demonstrates effective communication by transmitting real-time sensor data wirelessly to a mobile application at 5 Hz. It converts raw sensor readings into a clear CSV format and provides instant fault identification. Safety information is also communicated through visual indicators (LEDs) and audible alarms (buzzers).
PO(k): Project Management and Finance	The project offers a low-cost alternative to PLC-based railway systems by using an ESP32-WROOM-32 and affordable sensors. Efficient hardware planning, including proper GPIO selection to avoid pin conflicts, helped reduce development risks, costs, and implementation time.

i) Addressing Complex Engineering Problems (P)

Briefly explain how your project addresses P1 (Depth of knowledge), P2 (Range of conflicting requirements) and P4 (Familiarity of issues).

Attributes	Justification
P1 (Depth of knowledge)	The project requires knowledge of embedded systems, real-time programming, and fault-tolerant design. It uses a 3-out-of-4 voting algorithm, non-blocking timing with millis(), and reliable sensor interfacing on the ESP32.
P2 (Range of conflicting requirements)	The design balances fast safety response with false-alarm prevention. Sensor fusion and majority voting enable immediate train detection while using a confirmation period for fault detection without delaying gate operation.
P4 (Familiarity of issues)	The project addresses real-world challenges such as weather effects, sensor degradation, and environmental noise. Redundant ultrasonic and IR sensors ensure safe operation even when individual sensors fail.

j) Addressing Complex Engineering Activities (A)

Briefly explain how your project addresses A1 (Range of resources) and A2 (Level of interaction).

Attributes	Justification
A1 (Range of resources)	The project integrates multiple hardware and software resources, including ultrasonic and IR sensors, servo motors, LEDs, buzzers, ESP32 libraries, Bluetooth communication, and custom monitoring algorithms.
A2 (Level of interaction)	The system combines embedded control, mechanical actuation, and wireless monitoring. The ESP32 simultaneously manages sensors, controls the railway gate, and transmits real-time data to a mobile application for remote monitoring.

k) Conclusion and Future scope

Analysis of Objective Fulfillment

Objective 1 Fulfillment: The 3-out-of-4 voting algorithm was successfully implemented on the ESP32. Testing showed that the system responds quickly to train detection by activating the gate, warning LEDs, and buzzer. It also continued operating safely during single-sensor failure scenarios.

Objective 2 Fulfillment :The real-time monitoring system successfully detected and reported sensor faults. Fault information was transmitted to the mobile application via Bluetooth at a 5 Hz update rate, enabling effective remote monitoring.

Future Aspects and Research Directions

- Integrate LTE-M, NB-IoT, or LoRaWAN communication for long-range remote monitoring.
- Add solar power and battery management to support off-grid deployment.
- Incorporate camera-based computer vision to detect vehicles trapped on the crossing.
- Implement predictive maintenance algorithms to identify sensor degradation before failure occurs.

l) References

- [1] Y. Jiang, H. Low, and T. Clarke, "Design and Formal Verification of a 3-out-of-4 (3oo4) Majority Voting Architecture for Safety-Critical Railway Interlocking Control Systems," *IEEE Transactions on Reliability*, vol. 71, no. 2, pp. 642–655, Jun. 2022.
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- [3] M. Al-Rousan and O. Al-Araidah, "Edge-Computed Diagnostics and Real-Time Serial Telemetry Pipelines for Autonomous Grade Crossing Barriers Using 32-Bit Microcontrollers," *International Journal of Civil and Structural Engineering Automation*, vol. 14, no. 3, pp. 201–214, Oct. 2023.
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